



Components of computer system

- > computer is an operation-reduced system of 0s and 1s
 - zero describes the polarity of off, false, close, no, negative, and etc.
 - one describes the polarity of on, true, open, yes, positive, and etc.
 - the reduced system is known as the binary system (base-2) and it is used because it mirrors the electrical nature of computer hardware
- > it acts as a highly efficient machine that processes, stores, and communicates information using only two distinct states

Purpose of a computer

- > make work easier and faster

Computer System

- > consists of hardware and software components

Hardware

- > the physical equipment

Operating System (OS)

- > instructs the computer how to operate

Application software

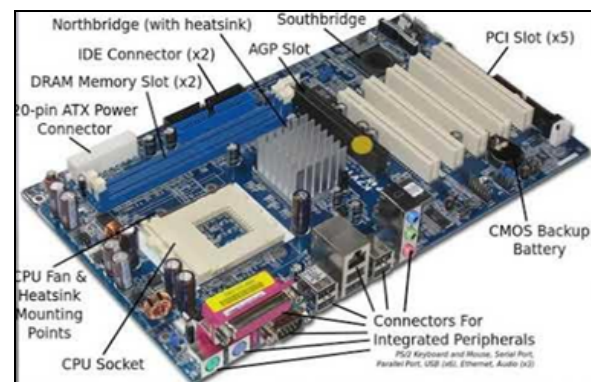
- > performs different functions or program with specific task

- Programs vary widely depending on the type of information that will be accessed or generated

Computer Hardware

Motherboard

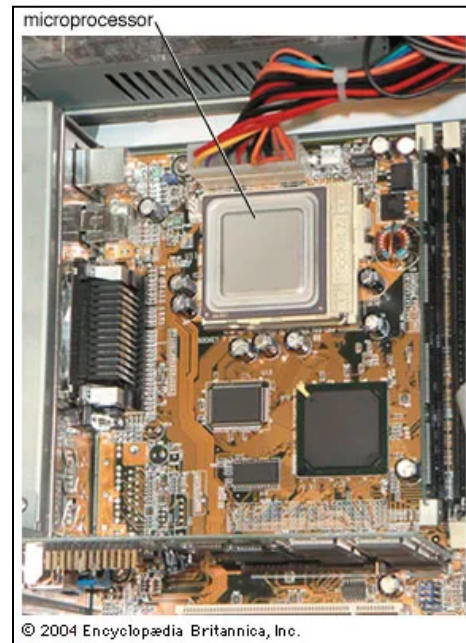
- > Main printed circuit board and contains the buses, or electrical pathways, found in a computer
 - Buses allow data to travel between the various components that comprise a computer.
- > Also known as the system, the backplane, or the main board



- > **CPU Socket** – Where the processor (CPU) is installed.
- > **CPU fan & heatsink mounts** – Hold the cooler that keeps the CPU from overheating.
- > **DRAM (RAM) slots** – Where memory sticks are placed.
- > **20-pin ATX power connector** – Main power supply connection for the motherboard.
- > **IDE connector** – Old port for hard drives and optical drives.
- > **AGP slot** – Old slot used for graphics cards.
- > **PCI slots** – Slots for expansion cards (sound card, network card, etc.).

- > **Northbridge** – Chip that handled fast components like CPU, RAM, and graphics (older boards).
- > **Southbridge** – Chip that handled slower parts like USB, audio, and storage (older boards).
- > **CMOS battery** – Keeps time and saves BIOS settings when the PC is off.
- > **Integrated peripherals connectors** – Ports for USB, audio, LAN, and other built-in features.
- > **Motherboard base (PCB)** – The main board that connects all components electrically.
- > **Chipset** – Controls how the CPU communicates with other parts.
- > **BIOS/UEFI chip** – Starts the computer and controls basic hardware settings.
- > **VRMs (power regulators)** – Control and stabilize power going to the CPU.
- > **SATA ports** – Connect modern hard drives and SSDs.
- > **M.2 slot** – Connects fast SSD storage.
- > **Rear I/O ports** – External ports (USB, HDMI, audio, LAN).
- > **Front-panel headers** – Connect the case's power button, reset button, and LEDs.

Central Processing Unit (CPU)



- > executes a program which is a sequence of stored instructions
- > each model of processor has an instruction set which it executes.
- > executes the program by processing each piece of data as directed by the program and the instruction set

3 Components of a CPU

ALU (Arithmetic Logic Unit) – Performs calculations and logical operations.

CU (Control Unit) – Directs and controls all CPU operations.

Memory – Stores data and instructions for processing.

Types of CPU

Reduced Instruction Set Computer (RISC)

- > Architectures use a relatively small set of instructions, and RISC chips are designed to execute these instructions very rapidly.

Complex Instruction Set Computer (CISC)

> Architectures use a broad set of instructions, resulting in fewer steps per operation.

Single Core CPU

> One core inside a single CPU chip that handles all of the processing capability

- A motherboard manufacturer may be able to build a powerful, multi-processor computer.

Dual Core CPU

> Two cores inside a single CPU chip in which both cores can process information at the same time

Overclocking

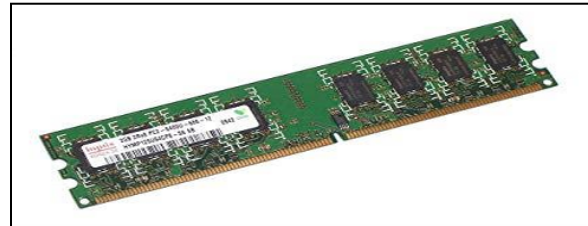
> a technique used to make a processor work at a faster speed than its original specification
> it is not a reliable way to improve computer performance and can result in damaging the CPU

Multi Media Extension (MMX)

> set of multimedia instructions built into intel processors

- MMX enabled microprocessors can handle many common multimedia operations that are normally handled by a separate sound or video card
- However, only software especially written to call MMX instructions can take advantage of the MMX instructions set

Random-Access Memory (RAM)



> It is the temporary storage for data and programs that are being accessed by the CPU
> it is volatile

Types of RAM

Dynamic Random Access Memory (DRAM)

> a memory chip that is used as main memory
> must be constantly refreshed with pulses of electricity in order to maintain the data stored within a chip.

Static Random Access Memory (SRAM)

> a memory chip that is used as cache memory
> much faster than DRAM and does not have to be refreshed as often.

Fast Page Mode DRAM (FPM Memory)

> memory that supports paging

- paging enables faster access to the data than regular DRAM

> most 486 and Pentium systems from 1995 and earlier use FPM memory

Extended Data Output RAM (EDO Memory)

> memory that overlaps consecutive data accesses
> speeds up the access time to retrieve data from memory because the CPU does not have to wait from one data

access cycle to end before another data access cycle begins

Synchronous DRAM (SDRAM)

> operates in synchronization with the memory bus

- The memory bus is the data path between the CPU and the main memory.

Double Data Rate 2 SDRAM (DDR2 SDRAM)

> faster than DDR SDRAM memory
> improves performance over DDR SDRAM by decreasing noise and crosstalk between the signal wires

Rambus DRAM (RDRAM)

> a memory chip that was developed to communicate at very high rates of speed
> RDRAM chips are not commonly used

Power Supply



> the piece of hardware that's used to convert the power provided from the outlet into usable power for the many parts inside the computer case

Video Card



> a piece of computer hardware that's numerous contacts on the bottom of the card and one or more ports on the side for connection to video displays and other devices

Storage Drives

> storage drive reads or writes information to magnetic or optical storage media.
> the drive can be used to store data permanently or to retrieve information from a media disk

Types of Storage Drives

Floppy Drive



> other name: floppy disk drive
> a storage device that uses removable 3.5-inch floppy disk

Hard Drive



- > other name: hard disk drive
- > a magnetic storage device that is installed inside the computer

Optical Drive



- > a storage device that uses lasers to read data on the optical media
- > There are two types of optical drives:
 - Compact Disk (CD)
 - Digital Versatile Disk (DVD)= media which can be pre-recorded (read-only), recordable (write once), or re-recordable (read and write multiple times)

Types of RAM

CD-ROM – CD reads only memory media that is pre-recorded.

CD-R – CD-recordable media that can be recorded once.

CD-RW – CD-rewritable media that can be recorded, erased, and recorded

DVD-ROM – DVD read-only memory media that is pre-recorded

DVD-RAM – DVD-random access memory media that can be recorded, erased, and re-recorded.

DVD+/-R – DVD-recordable media that can be recorded once.

DVD+/-RW – DVD-rewritable media that can be recorded, erased, and re-recorded

Flash Drive



- > other name: thumb drive
- > a removable storage device that connects to a USB port.
- > uses a special type of memory that requires no power to maintain the data

Note: Hard drives and optical drives are manufactured with different interfaces that are used to connect the drive to the computer. To install a storage drive in a computer, the connection interface on the drive must be the same as the controller on the motherboard

Types of Drive Interfaces

Integrated Drive Electronics (IDE)

> other name: Advanced Technology Attachment (ATA)

> an early drive controller interface that connects computers and hard disk drives

> an IDE interface uses a 40-pin connector

Enhanced Integrated Drive

Electronics(EIDE)

- > other name : ATA-2
- > an updated version of the IDE drive controller interface
- > supports hard drives larger than 512MB, enables Direct Memory Access (DMA) for speed, and uses the AT Attachment Packet interface (ATAPI) to accommodate optical drives and tape drives on the EIDE bus
- > an EIDE interface uses a 40-pin connector.

Parallel ATA (PATA)

- > refers to the parallel version of the ATA drive controller interface

Serial ATA (SATA)

- > refers to the serial version of the ATA drive controller interface
- > a SATA interface uses a 7-pin connector

Small Computer System Interface (SCSI)

- > a drive controller interface that can connect up to 15 drives
- > connects both internal and external drives
- > an SCSI interface uses a 50-pin, 68-pin, or 80 pin connector

Ports and Cables

Serial Ports and Cables

- > can be either a DB-9 or a DB-25 male connector
- > transmits one bit of data at a time.

USB Ports and Cables

- > the Universal Serial Bus (USB) is a standard interface that connects peripheral devices to a computer

FireWire Ports and Cables

- > a high-speed, hot-swappable interface that connects peripheral devices to a computer

Parallel Ports and Cables

- > a parallel port on a computer is a standard Type A DB-25 female connector, The parallel connector on a printer may use a Type C high-density 36-pin connector

SCSI Ports and Cables

- > it can transmit data at rates in excess of 320Mbps and can support up to 15 devices
- > if a single SCSI device is connected to an SCSI port, the cable can be up to 80 feet (24.4m) in length.
- > Types of it:
 - DB-25-female connector
 - High-density-50-pin female connector
 - High-density 68-pin female connector

Network Ports and Cables

- > other name: RJ-45 port
- > connects a computer to a network
- > the connection speed depends on the type of network port

PS/2 Ports

- > connects a keyboard or a mouse to a computer
- > is a 6 pin mini-DIN female connector

Audio Ports

- > connects audio devices to the computer

Line In

- > connects to an external source, such as a stereo system.

Microphone

> connects to a microphone

Line Out

> connects to speakers or headphones

Game port/MIDI

> connects to a joystick or MIDI-interfaced device

Video Ports and Connectors

> connects a monitor cable to a computer

> There are several video port and connector types:

- **Video Graphics Array (VGA):** VGA has a 3-row 15-pin female connector and provides analog output to a monitor.
- **High Definition Multimedia Interface (HDMI):** HDMI has a 19-pin connector and provides digital video and digital audio signals
- **Component/RGB:** RGB has three shielded cables (red, green, blue) with RCA jacks and provides analog video signals.



Operating System and Application Software

Operating System (OS)

> the most important program that runs on a computer
> computer operating systems perform basic tasks such as:

- recognizing input from the keyboard
- sending output to the display
- keeping files and directories on the storage drives
- controlling peripheral devices such as printers

> Almost all modern operating systems can support more than one user, task, or CPU

4 main roles of OS

> Manage hardware
> Run applications
> Provide a user interface
> Manage files

Purpose of an OS

> The operating system boots the computer and sets up the file system

2 types of OS interfaces

> **Command Line Interface (CLI):** The user types commands at a prompt.
> **Graphical User Interface (GUI):** The user interacts with menus and icons

OS functions

> **File and folder management** - The operating system creates a file structure on the hard disk to drive to allow data to be stored
> **Application management** - The operating system manages all programs to ensure that the correct resources are allocated to the applications.
> **Control hardware access** - The operating system handles the interaction between the applications and hardware.

> **User interface** - The operating system enable the user to interact with software and hardware

Plug and Play Technology (PnP)

> The operating system is designed to automatically discover new plug-and-play (PnP) compatible hardware, and then configure the device, OS, and registry

Applications Programming Interface (API)

> allows programmers to create software applications consistent with the operating system

- **Open Graphics Library (OpenGL):** Cross-platform standard specification for multimedia graphics.
- **DirectX:** Collection of API's related to multimedia tasks for Microsoft Windows

Multi-user

> A type of operating system that allows multiple users to access the operating system at the same time.

Multitasking

> A method a computer uses to share resources between multiple tasks, or process.

Multiprocessing

> A type of computer that can support more than one physical processor or the ability to schedule tasks between multiple CPUs.

Multithreading

> A method used by a CPU to process multiple threads, or tasks within a program, at the same time

Computer System

> an integrated set of hardware, software, data, and users that work together to input, process, store, and output data

- in here an operating system (OS) is the essential, core software component of a computer system that manages hardware (CPU, memory, storage) and software resources. It acts as a bridge, enabling interaction between the user, application programs, and physical hardware
- > they can operate on either the two:
 - **Desktop Operating System (DOS):** intended for use in a small office/home office (SOHO) with a limited number of users
 - **Characteristics:**
 - Designed for single users
 - Designed to run personal applications
 - Designed for ease of use and user interaction
 - Designed for standalone computers
 - Designed with basic security features
 - **Common tasks:**
 - Running applications (word

- processing, browsers, games, media players)
- File management (create, save, delete, organize files)
- Device management (printers, USB, speakers, cameras)
- User interface management (windows, icons, menus)
- Internet access (web browsing, email, downloads)
- Supports personal productivity and home/school use
- Commonly used DOS fall into three groups:
 - **Windows:** dominates the personal computer world, offering a graphical user interface (GUI), virtual memory management, multitasking, and support for many peripheral devices
 - **Mac:** Mac OS features a graphical user interface (GUI) that utilizes windows, icons, and all applications that run on a Macintosh computer have a

- similar user interface.
- **Linux:** a freely distributed open source operating system that runs on a number of hardware platforms
- In these 3 groups OS share several similarities with it:
 - GUI Interface
 - Standard “look and feel” characteristics
 - Multithreading capabilities
 - Multitasking capabilities
 - Ability to function with most hardware
- Differences between desktop operating systems are typically related to availability and how much can be accomplished using the GUI:
 - **Windows and MAC OS** users can perform the majority of tasks through the GUI.
 - **Linux and UNIX** users most understand the use of CLI to perform some tasks
- Network Operating System (NOS):
 - designed for a corporate environment serving multiple users with a wide range of needs

- **Characteristics:**

- Designed to support multiple users
- Designed to run multi-user applications
- Designed to be robust and redundant
- Designed to be used on a network
- Designed with increased security compared to desktop operating systems

- **Common tasks:**

- File Transfer Protocol (FTP)
- Simple Mail Transfer Protocol (SMTP)
- Hypertext Transfer Protocol (HTTP)
- Lightweight Directory Access Protocol (LDAP)
- Active Directory
- designed to function well in a client/server environment
- A server is a robust computer that is used to service the needs of multiple clients within a network

- **Attributes:**

- has much of the same functionality

as a desktop operating system

- Determining the number of users that a server can support depends on factors such as the hardware specifications, the network operating system, and physical demands on the system.
- will normally remain stable when the number of users is high, but the process may become slow

- **Common NOS:**

- Novell Netware
- Microsoft Windows Server
- Linux
- UNIX

> these are the characteristics of a computer system:

- Designed to support a single user
- Designed to run single-user applications
- Designed to share files and folders on a small network
- Designed to share peripherals on a small network

Code for an OS

Open source applications

> can be read and modified

> programmers openly share code with other programmers

- Ex: Linux distributions are open source

Proprietary application

- > cannot be read or modified
- > proprietary software agreements restrict the use of the software, identifying where and when the software may be used

Determine OS based on customer needs

- > Select the proper operating system to meet the needs of your customer, gain as much information about the customer's daily activities as possible.
- > Once you have a thorough understanding of what the customer needs, you can successfully select appropriate software and hardware to satisfy existing and future requirements

Determine OS based on customer requirements

- > What Does Your Customer Require?
 - To identify applications that customers will use and ensure compatibility
- > When selecting hardware and software, the needs of your customer should come first.
- > Remain neutral and listen to the needs of your customer before deciding on any computer hardware or software solution.

Explore customer needs by asking the following types of questions

- > What general office applications, such as word processing, spreadsheets, or presentation software, does your customer require?
- > What graphics, such as Photoshop or Illustrator, do your customer require?

> What animation, such as Flash, does your customer require?

> What business applications, such as accounting, contact management, sales tracking or database, does your customer require?

Last step

> Once you have determined the types of software your customer requires, you should determine which available products will meet these needs.

> Before recommending applications, you should review the existing operating systems and software applications that your customer is using.

> The following are some of the factors to consider:

- Is compatibility an issue?
- Which operating system is required for current applications?
- Will the applications work with the existing hardware?
- Do any files have to be transferred between systems?
- Is there an issue with the format of the file systems involved?
- Are the applications standalone or networked?

Identify Minimum Hardware Requirements

> Operating systems and applications have minimum hardware requirements that must be met for the computer to be functional.

> In some cases, the application requirements may exceed the requirements of the operating system.

- > For the application to function properly, it will be necessary to satisfy any additional requirement.
- > Meeting only the minimum requirements may not be beneficial to your customer in the long term. Your customer may need to upgrade or purchase additional hardware
- > Increasing the following are some common upgrades to the minimum requirements:
 - RAM capacity
 - Hard drive size
 - Processor speed
 - Video card memory and speed
- > Identify the equipment that your customer has in place when you are analyzing hardware requirements.
- > If upgrades are required to meet minimum requirements, conduct a cost analysis to determine the best course of action.
 - In some cases, it may be necessary for the customer to purchase new equipment.
 - In other cases, it may be cost-effective to upgrade the existing equipment.

Hardware Compatibility List (HCL)

- > Most of operating systems have a hardware compatibility list that can be found on the manufacturer's website.
- > These lists provide a detailed inventory of hardware that has been tested and is known to work with the operating system.
- > If the hardware already in place is not on the list, the hardware may require upgrading. If a hardware component is

not on the list, there may be problems once installed

Application Software

- > user-oriented, end-user program designed to perform specific, productive tasks—such as word processing, web browsing, or data management—rather than managing the computer itself.
- > These apps, including Microsoft Word, Google Chrome, and mobile tools, run on top of system software (OS) and can be customized or general-purpose.



Computer Hardware Input, Process, Output, Storage, and Network

Computer Hardware

- > refers to the physical equipment or components of an electronic data processing. It is also composed of input, output and storage and network devices

Input Devices

- > they are the hardware devices which take information from user of the computer system, convert it into electrical signals and transmit it to the processor

Keyboard

- > A computer hardware that used to input text characters and other commands

Mouse

> A device that used to point to and select items on your computer screen

Image Scanner

> A Device that used to transfer images or text a computer there are special models the negatives or to scan book

Web camera

> It is low resolution video camera used to provide visual input

Microphone

> an input device to input sound that is then stored in a digital form. This is used to add sounds

Output Devices

> used to present information to the user from a computer.

> Output devices take data from the computer system and convert it to a form that can be read by humans

Monitor

> The most important difference between these monitor types is the technology used to create an image:

- **Cathode-ray (CRT):** The most common monitor type, red, green, blue electron beams. Most tv's used this technology



- **Liquid crystal display (LCD):** commonly used in laptops and

some projectors. It consist of two polarizing filters with a liquid crystal solution



Digital Light Processing (DLP)

> a technology used in projectors, DLP projectors use a spinning color wheel with a microprocessor controlled array of mirrors called a digital micromirror device (DMD)



Printers and Fax Machines

> Printers are output devices that create hard copies of computer files. Some printers specialize in particular applications

Speakers and Headphones

> they are output devices for audio

Storage devices

> other names:

- digital storage
- Storage
- storage media
- storage medium

> any hardware capable of holding information either temporarily or permanently

Magnetic storage device

> one of the most common types of storage used with computers



Optical storage device

> this uses lasers and lights as its method of writing and reading data



Flash memory

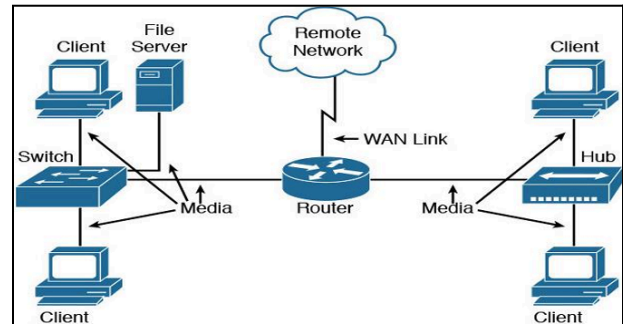
> Has started to replace magnetic media
> it is more efficient and reliable

Paper storage

> Early computers had no method of using any of the above technologies for storing information, these form is rarely used



Network Devices



Explanation: This diagram illustrates how data moves in a client-server network across local and remote networks. Clients are connected to networking devices using physical media such as Ethernet cables: a switch on the left, which sends data only to the intended device, and a hub on the right, which broadcasts data to all connected clients. When a client requests a file or service, the data first passes through the switch or hub, then goes to the router, which analyzes the destination address. If the data is meant for another local device, it is delivered directly; if it is for a remote network, the router forwards it through the WAN link to the remote network, and the response follows the same path back to the client.

Repeater

> operates at the physical layer
> Its job is to regenerate the signal over the same network

HUB

> a multiport repeater
> connects multiple wires coming from different branches
> there are two type:

- **Active HUB** — have their own power supply and can clean,

boost and relay the signal along the network

- **Passive HUB** — collects wiring from nodes and power supply from active hub. These hub relay signals onto network without cleaning and boosting

Bridge

- > operates a data link layer
- > it is a repeater, with add on functionality of filtering content by reading the MAC addresses of source and destination
- > used for interconnecting two LANS working on the same protocol
- > there are two types:
 - **Transparent Bridge** — stations are completely unaware of the bridges existence i.e
 - **Source Routing Bridge** — used on token ring networks, and is standardized in Section 9 of the IEEE 802.2 standard. It is simpler (spanning tree protocol is not necessary) and much of the bridging functions are performed by the end systems, particularly the sources, giving rise to its name

Switch

- > a multi port bridge with a buffer and a design that can boost can perform error, it divides collision domain of host

Routers

- > In order for devices on the network to connect to the Internet, the router must be connected to a modem
 - Therefore, most routers have a specific Ethernet port that is

designed to connect to the Ethernet port

Gateway

- > a passage to connect two networks together that may work upon different networking models

Router

- > other name: bridging router
- > combines features of both bridge and router
- > It is working as a bridge
- > it can filter local area network traffic

Note: DISCLAIMER all info is galing ppt ni madam di ako sure kung tama pagka ayos ko sa info from there pero basta galing ppt un. Ung ibang info is galing google, soafer onti sa recording ni madam(meron nmn sa drive plus tinamad ako mag take note ng seryoso HAHAAAAHA). UNG IBANG INFO IS SINASABI NYA kaya parang kulang sa ppt nya. RECOMMEND KO na mag podcast ka and sariling note kasi this is 90% ppt ni mam HAHAAAAHA un lng pasabi if may mali un lng ty rev well mga gais<3